Introduction: System Design Patterns

This lesson gives a brief overview of the system design patterns that we will be discussing in the following lessons.

In the following chapters, we will discuss a set of system design patterns. These patterns refer to common design problems related to distributed systems and their solutions. Knowing these patterns is very important as they can be applied to all types of distributed systems and are very handy, especially in a system design interview.

Here is the list of patterns we will be discussing:

1. Bloom Filters
2. Consistent Hashing
3. Quorum
4. Leader and Follower
5. Write-ahead Log
6. Segmented Log
7. High-Water mark
8. Lease
9. Heartbeat
10. Gossip Protocol
11. Phi Accrual Failure Detection
12. Split-brain
13. Fencing
14. Checksum
15. Vector Clocks
16. CAP Theorem
17. PACELEC Theorem
18. Hinted Handoff
19. Read Repair
20. Merkle Trees

Let's get going.